

FITBUDDY

AI Fitness Plan Generator using Gemini Models

PROJECT DOCUMENTATION

Team Details	Information
Team	Team 1D
Team ID	SWTID-2026-3884
Team Size	5
Leader	Karthika V
Leader Email	karthikakanishkan@gamil.com
Team Member	Kaviya Anandhan — kskaviyask142@gmail.com
Team Member	G Kaviya — Vkaviya9507@gmail.com
Team Member	P Keerthana — keerthanamegala2008@gmail.com
Team Member	Kiruthika A — arumugamrasu335@gamil.com

Technology Stack: FastAPI • Jinja2 • SQLite • SQLAlchemy • Google Gemini

1. Project Overview

FitBuddy is a FastAPI + Jinja2 + SQLite application that generates personalized 7-day workout plans and nutrition/recovery tips using Google Gemini. Users can submit feedback to regenerate a plan, while an admin page can view and delete stored users.

2. Features

- User form: name, user ID, age, weight, goal, and intensity.
- AI-generated 7-day workout plan.
- Nutrition and recovery tip.
- Feedback-based plan update.
- SQLite persistence using SQLAlchemy.
- Admin view of users and original/updated plans.
- FastAPI Swagger documentation at /docs.
- DEMO_MODE for testing without an API key.

3. Objectives

- Generate personalized workout plans.
- Provide a structured 7-day schedule.
- Provide nutrition and recovery guidance.
- Allow feedback-based plan regeneration.
- Persist user and plan information.
- Provide an admin interface.
- Support testing through Demo Mode.

4. Technologies Used

Technology	Purpose
Python	Backend programming
FastAPI	Web/API framework
Jinja2	HTML template rendering
SQLite	Database
SQLAlchemy	Database ORM
Google Gemini	AI plan generation
Uvicorn	Application server
HTML/CSS	Frontend and styling
python-dotenv	Environment configuration

5. Project Structure

FitBuddy_AI_Fitness_Plan_Generator/

- app/
- ■■■■ main.py
- ■■■■ routes.py
- ■■■■ database.py
- ■■■■ models.py
- ■■■■ schemas.py
- ■■■■ gemini_generator.py
- ■■■■ gemini_flash_generator.py
- ■■■■ updated_plan.py
- ■■■■ templates/
- ■■■■ static/
- data/
- .env.example
- .gitignore
- requirements.txt
- README.md

6. Module Description

main.py — FastAPI entry point.

routes.py — Application routes and request handling.

database.py — SQLite and SQLAlchemy configuration.

models.py — Database models.

schemas.py — Request/response structures.

gemini_generator.py — Gemini workout generation.

gemini_flash_generator.py — Gemini Flash generation.

updated_plan.py — Feedback-based updated plan generation.

templates/ — Jinja2 HTML pages.

static/style.css — UI styling.

7. Application Workflow

User opens application → enters fitness details → FastAPI validates input → Gemini generates a 7-day workout plan and nutrition/recovery tip → data is stored in SQLite → result is displayed → user submits feedback → updated plan is generated.

8. Main Routes

Method	Endpoint	Purpose
GET	/	Home page
POST	/generate-workout	Generate workout
POST	/submit-feedback	Update plan using feedback
GET	/view-all-users	Admin user view

Method	Endpoint	Purpose
POST	/delete-user/{user_id}	Delete user

9. Database

SQLite with SQLAlchemy is used for persistence of user information and generated plans.

10. Gemini AI Integration

The application uses the modern google-genai SDK. API configuration is kept in the environment file and model names are configurable.

```
GOOGLE_API_KEY=your_api_key
DEMO_MODE=false
```

11. Demo Mode

With DEMO_MODE=true, the application can be tested without Gemini and returns deterministic sample plans and tips.

12. Installation and Running

Open PowerShell in the project folder:

```
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
Copy-Item .env.example .env
uvicorn app.main:app --reload
```

13. Application URLs

Website: <http://127.0.0.1:8000>

API Docs: <http://127.0.0.1:8000/docs>

Admin View: <http://127.0.0.1:8000/view-all-users>

14. Advantages

- Personalized workout generation
- AI-powered recommendations
- 7-day structured plan
- Feedback-based regeneration
- Persistent database storage
- Admin user management
- Swagger API documentation
- Demo mode for testing

15. Limitations

- AI-generated fitness plans may require professional review.
- Gemini API/model availability depends on configuration.
- Demo Mode provides sample output.
- This is an educational application, not a substitute for medical or professional fitness advice.

16. Future Enhancements

- User authentication
- Progress tracking
- BMI and calorie calculation
- Exercise video integration
- Mobile-responsive UI
- Admin authentication
- PDF workout-plan export
- Workout reminders

17. Conclusion

FitBuddy combines FastAPI, SQLite, SQLAlchemy, Jinja2 and Google Gemini to provide an AI-powered fitness planning system. Users receive personalized 7-day workout plans and nutrition/recovery tips, can submit feedback, and can receive updated plans.

18. Safety Note

AI output is not a substitute for medical or professional fitness advice. Users should adjust activity to their abilities and seek professional guidance where appropriate.